

Data Center Power Supply Systems: From Grid Edge to Point-of-Load

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Abstract—Data centers consume about 2% of the world's electricity with continuing growth. The power supply system plays a significant role in the energy saving and decarbonization of data centers. The development of power electronics brings opportunities for more efficient and reliable data centers. This article presents an overview of the data center power supply system covering the power delivery path from the grid edge to onboard point-of-load (PoL) conversion. The system architectures are introduced at first with the discussion on efficiency and reliability. The power conversion stages of data center power supply system are discussed as ac–dc conversions and dc–dc conversions. The state-of-the-art techniques in topology, control, and device are investigated. This article is an attempt to provide an overview for high-performance data center power supply system design.

Index Terms—ac–dc, data center, dc–dc, efficiency, reliability, soft-switching, solid state transformer (SST), switched-capacitor (SC), uninterruptible power supply (UPS).

I. INTRODUCTION

THE explosive demand for data and computing and the development of 5G technology have driven the rapid growth of data centers. It is reported that the global data center market has exceeded 200 billion U.S. dollars in 2021 [1], and is expected to increase with a compound annual growth rate around 5%–10% in the next decade [2], [3]. As a result, the energy consumption of data centers also continues to grow. The annual energy consumption of global data center (including data computing and transmission) in 2021 is estimated about 480–660 TWh [4], accounting for 1.7%–2.2% of the world's electricity generation [5]. Such high-energy consumption of data centers has raised public concerns about their economic and environmental impact.

Manuscript received 14 July 2022; revised 26 September 2022 and 17 November 2022; accepted 6 December 2022. Date of publication 14 December 2022; date of current version 13 June 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 52037010 and in part by the Zhejiang Provincial Natural Science Foundation of China under Grant LQ23E070004. Recommended for publication by Associate Editor Joseph O. Ojo. (*Corresponding author: Dehong Xu.*)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JESTPE.2022.3229063>.

Digital Object Identifier 10.1109/JESTPE.2022.3229063

Electricity usually accounts for 40%–80% of the long-term operating expenditures of a data center [6]. Improving the energy efficiency of data centers can directly lead to significant economic benefits. Data center owners/operators care about one metric called power usage effectiveness (PUE), which is the ratio of a data center's total electricity usage to the electricity used by IT equipment. The average annual PUE for a large-scale data center was 1.57 in 2021 [7] and the best-in-class PUE has dropped to 1.1 [8] by optimizing the power distribution path and improving the efficiency of the uninterruptible power supply (UPS) and cooling system. Besides PUE, the power supply units (PSUs) and point-of-load (PoL) converters, which deliver power from the rack to end loads such as central processing unit (CPU), graphic processing unit (GPU) and storage, are also critical to the overall energy efficiency of data centers. Therefore, the entire power conversion path should be investigated to drive the efficiency improvement of the entire data center power supply system.

The data center industry is considered to have high carbon emission. Following the global trend toward net-zero carbon, the data center industry is also striving to achieve the goal of zero carbon emissions by employing clean energies such as solar, wind, and hydrogen fuel. Google, Apple, and Facebook have announced that their operational energy usage, including the data centers, is fully supplied by renewable energy. To replace diesel generators, Microsoft has tested hydrogen fuel cells as backup power in data centers [9]. As a result, data center power supply systems need to be upgraded to interface with renewable energy.

The power supply system is also critical to the reliability of a data center. An industry survey shows that power is the leading cause of data center outage (43%) [7]. The reliability can be enhanced at all levels of the power supply system: new system architectures to reduce the conversion stages, new circuit topologies and control strategies to reduce electrical and thermal stress, and new devices with better performance.

Power electronics for data centers continues to evolve to address these challenges. In [10], the data center challenges and the related power electronics are reviewed from the utility level to the internal-chip level. This article focuses on the emerging technologies of data center power supply system in recent years, covering the ac interface at the grid edge to PoL converters. The architectures of data center power supply systems are introduced in Section II. The power conversion stages in data center power supply systems are divided into ac–dc conversions and dc–dc conversions. Section III intro-

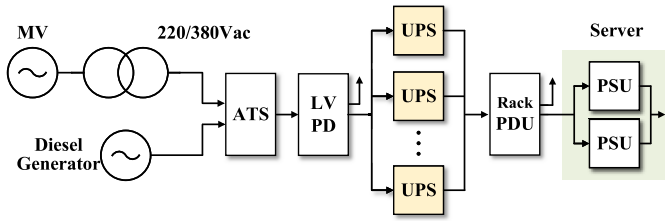


Fig. 1. Typical data center power supply system from the MV grid to the server PSU.

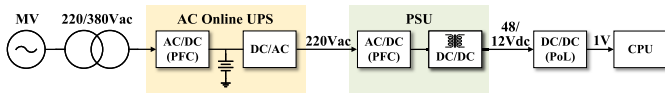


Fig. 2. Data center power supply system with ac online UPS (ac UPS system).

II. SYSTEM ARCHITECTURE

This section provides an architecture-level review of data center power supply systems. Fig. 1 shows a typical data center power supply system from the medium voltage (MV) grid to the server PSU [10]. The utility grid is the main power source of the data center. Besides, diesel generators are the backup power source for servers and the cooling system. They are interconnected by the automatic transfer switch (ATS). Then the power is delivered to the server by UPS and PSU with parallel redundancy for reliability. The power supply systems are categorized as ac architectures, dc architectures, and hybrid architectures. For simplicity, the following discussion only focuses on the power electronics converters of data center power supply systems. The diesel generator, power distribution units (PDUs), protection, and redundancy are not included.

A. AC Power Architecture

The ac power architecture has existed in data centers over decades. As shown in Fig. 2, the ac power architecture consists of UPS, PSU, and POL converter to CPU. Power is distributed in ac at both building-level and rack-level. UPS is critical to the reliability, power quality, and stability of the entire system. The classification, topology, and control of UPS are comprehensively reviewed in [11] and [12]. Among them, the online UPS has better tolerance to the grid variation, precise output regulation, and seamless transition to backup, and has been widely used in data center power supply systems. Considering the power level of typical data center racks, the ac online UPS in data centers employs a three-phase ac–dc stage for power factor correction (PFC) and a three-phase dc–ac stage as a three-phase back-to-back (BTB) converter. The backup batteries are deployed on the dc bus of the BTB converter. The bypass switch of the ac online UPS is not

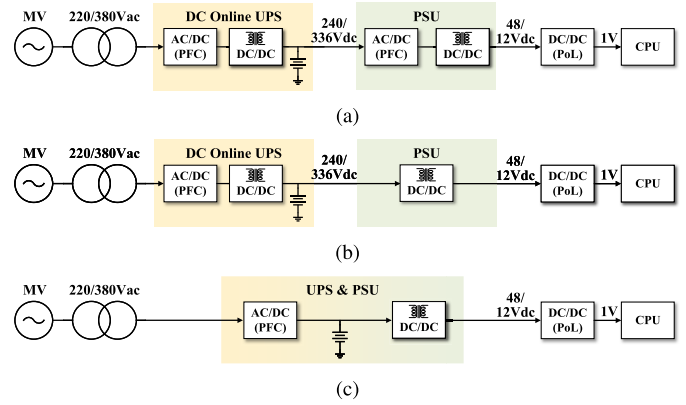


Fig. 3. Three dc power supply architectures. (a) Conventional HVDC architecture. (b) HVDC architecture with dc PSU. (c) Simplified LVDC architecture.

shown in Fig. 2. The double-conversion (ac–dc–ac) efficiency of commercial high-power ac online UPS is usually around 96%–97% [13], [14]. With silicon carbide (SiC) and three-level technologies, the double-conversion efficiency is higher than 98% at 50% load [15], [16], and the efficiency of each stage can reach 99% [17], [18].

The PSU also has two conversion stages: usually a boost-type PFC stage with a 400-V output and an isolated dc–dc stage converting the 400-V bus to a 12-V or 48-V bus for the motherboard power supply. The 80-Plus Titanium certification requires a 96% half-load efficiency and a 91% full-load efficiency for a PSU with the 230-Vac input [19]. The latest OCP open rack standard already pushes these metrics to 97.5% and 96.5% [20]. The last stage is the onboard PoL converter for the CPU power supply.

In the ac power architecture, there are six conversion stages in total from the grid to CPUs. The battery power also needs to be converted by the dc–ac stage in the ac UPS. Such long conversion path may decrease the overall efficiency and reliability of the power supply system. On the other hand, the ac power distribution is mature and inexpensive.

B. DC Power Architecture

Comparing to ac, paralleling in dc is simple and reliable, making dc power architectures attractive for data centers. Fig. 3(a) shows the power supply system with dc online UPS, which is also called the high voltage direct current (HVDC) system. In the HVDC system, ac power is distributed at building-level and dc power is distributed at rack-level. The ac online UPS is replaced by the dc online UPS. The first stage of the dc online UPS is the same as the three-phase PFC converter in ac online UPS. A second isolated dc–dc stage which typically employs LLC topology is needed to provide the 240-V or 336-V output voltage (nominal value, voltage range is 204–288 V and 300–400 V). High-voltage battery strings are directly connected to the HVDC bus. Except the UPS, other conversion stages are the same as the ac UPS architecture. ac PSUs are compatible with the HVDC system without modification.

Above two solutions are the mainstream data center power supply systems, both with online UPS and six conversion

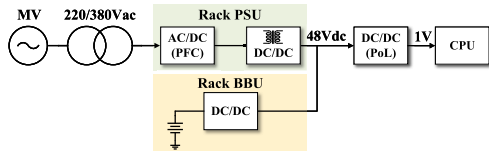


Fig. 4. 48-V rack power supply system with 48-V BBU [21].

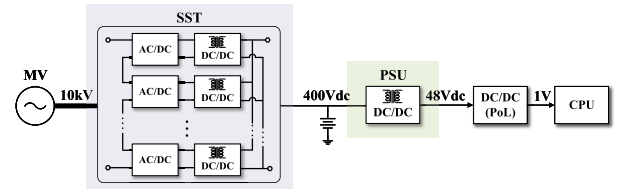


Fig. 6. Data center power supply system with SST.

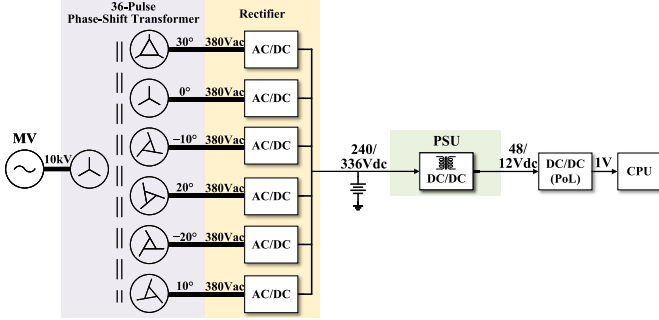


Fig. 5. Panama power supply system [23].

stages. In fact, the ac–dc stage of PSU is not necessary for the HVDC architecture and it can be removed as shown in Fig. 3(b). Fig. 3(c) shows a further simplified dc architecture, in which the batteries are deployed at the HVDC bus of the ac PSU and the power is distributed at low voltage (48 V or 12 V) among servers. Generally, dc architectures with fewer conversion stages can achieve higher efficiency. The dc power distribution also has flexible battery deployment with fewer conversion stages to the load, so that the efficiency of the dc system is higher in the backup mode and the related reliability is also higher. However, the dc breaker and protection for HVDC system are difficult and costly.

Google proposed the 48-V rack power supply system [21]. As shown in Fig. 4, both the PSU and the battery backup unit (BBU) are deployed on the rack shelf. The bidirectional dc–dc converter in the BBU is a four-switch buck-boost converter whose peak efficiency is above 98% [22]. Power is distributed at 48 V to servers with 16 times reduction of cable loss comparing to the conventional 12-V system. The overall efficiency is also improved with the reduction of total conversion stages. The challenges are that the PoL converters have to operate at higher input voltage and the backup battery capacity is limited by the volume of the rack shelf.

Recently, power electronics manufacturers together with data center owners have also focused on innovations in the grid interface stage up to MV level. Fig. 5 shows a data center power supply system called the Panama power supply [23]. The 36-pulse phase shift transformer connects the 10-kV grid and the multipulse ac–dc rectification system [24]. Each rectifier includes a diode rectifier bridge and a buck converter to regulate the output dc voltage. The 240-V or 336-V HVDC bus is also compatible with the universal ac PSUs. The case study in [23] shows the efficiency of the 36-pulse phase shift transformer is 99% and the peak efficiency of the rectifier is 98.5% with Silicon devices, resulting a 97.5% efficiency from the MV grid to the HVDC bus of a 2.5-MW data enter. Another recent study compares a similar 12-pulse rectifier with

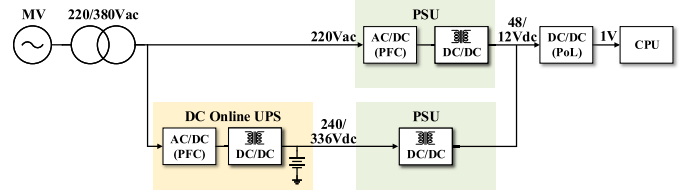


Fig. 7. ac + dc hybrid system with direct utility supply and dc UPS.

traditional line-frequency transformer (LFT)-based rectifiers and solid-state transformer (SST) [25]. The result shows the 12-pulse rectifier solution has the highest efficiency. Besides the efficiency, the Panama architecture also has high-power density and low cost because the circuit topology of the rectifier is greatly simplified.

The SST has raised the interest from the academia and industry [26], [27], [28], [29], [30], [31], [32], [33], recently. Fig. 6 shows the data center power supply with SST. Modular SST cells are configured as input-series output-parallel (ISOP) for interfacing the MV grid and the dc bus. In SST, energy is processed by semiconductor devices at high frequency rather than by transformer at line frequency. Thereby, the power density of the power supply system is increased. The full-load efficiency of a 3.8-kVac to 400-Vdc/25-kW laboratory prototype with 10-kV SiC device achieves 98.1% [29], [30]. And a 13.2-kVac to 1.05-kVdc/400-kW demonstration from industry shows a 98.5% efficiency around half-load [32]. The major advantage of SST is the high-power density. Furthermore, the HVDC output of SST is directly fed to server racks and the PSU only performs single-stage conversion. That also improves the overall efficiency.

C. Hybrid Power Architecture

Most top-tier data centers require dual power feeds for better redundancy. That is, there are two independent power distribution paths from the utility grid to servers. The most typical way is duplicating the power supply system in Figs. 2 and 3. But the pain is the cost of infrastructure and equipment. Fig. 7 shows an hybrid power supply system with direct utility supply and dc UPS. Power is equally shared during normal operation and the efficiency is improved owing to removing the online UPS in the ac path. The investment in UPS is also reduced. However, the ac path is not protected by UPS. Thereby, the reliability of this ac + dc hybrid system is lower than both the dual feeds systems.

More data centers are also using renewable energy to reduce the carbon emission. A super-UPS concept [34] is illustrated in Fig. 8, which is the evolution of UPS by adding natural gas, PV and Hydrogen fuel cell to the dc bus. The utility

TABLE I
COMPARISON OF DATA CENTER POWER SUPPLY ARCHITECTURES

Archit.	No. of Stages [*]	Normal Eff. [†]	Backup Eff. [‡]	Distribution Voltage		Battery Deploy	Remarks
				Building	Rack		
AC UPS Fig. 2	6, 4	93.15%	95.55%	220/380 Vac	220 Vac	Cent.	Mature, relatively low efficiency
HVDC Fig. 3(a)	6, 3	93.15%	97%	240/336 Vdc	240/336 Vdc	Cent.	Reliable, relatively low efficiency, higher cable loss at building level, and high cost on HVDC distribution
HVDC Fig. 3(b)	5, 2	94.11%	98%	240/336 Vdc	240/336 Vdc	Cent.	Reliable, efficiency higher than conventional HVDC, higher cable loss at building level, and high cost on HVDC distribution
LVDC Fig. 3(c)	4, 2	96.03%	98%	220/380 Vac	48 Vdc	Distrib.	Reliable, most efficient conversion, higher cable loss at rack level, requires deeply customized PSU and better maintenance
48-V Fig. 4	4, 2	96.03%	98%	220/380 Vac	48 Vdc	Distrib.	Reliable, most efficient conversion, higher cable loss at rack level, and difficulty in maintenance due to distributed batteries
Panama Fig. 5	4, 2	95.55%	98%	240/336 Vdc	240/336 Vdc	Cent.	Efficient, compact and low cost, higher cable loss at building level, need attention to the grid current harmonics
SST Fig. 6	4, 2	95.55%	98%	400 Vdc	400 Vdc	Cent.	Efficient, expected to be compact, lower cable loss, cost on HVDC distribution and semiconductor devices
Hybrid Fig. 7	4/5, 2	95.07%	98%	220/380 Vac 240/336 Vdc	220 Vac 240/336 Vdc	Cent.	Relatively high efficiency, lower reliability due to the lack of AC online UPS, moderate cable loss
Super-UPS Fig. 8	6, 5	93.15%	95.55%	220/380 Vac ± 375 Vdc	220 Vac	Cent.	Highest reliability, relatively low efficiency, easy to connect renewable sources

* From MV grid to CPU.

† From MV grid to the input of PoL converter.

‡ From Battery to the input of PoL converter.

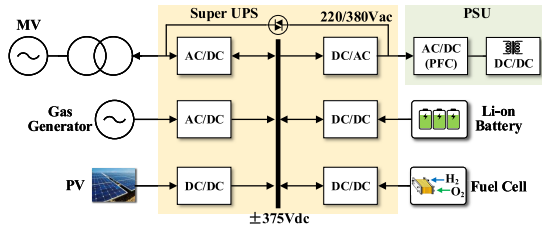


Fig. 8. Concept of Super-UPS [34].

grid and the natural gas are two major independent power sources for the infrastructure of the modern society, which are competing and naturally backing up each other. Fuel cell and Li-on battery support the backup power to the load when both the utility grid and gas pipelines fail. This architecture not only reduce the carbon emission but also increases the system reliability significantly. Its mean time between failures (MTBF) is 2.5 times of the traditional UPS [34]. The power management and fault protection are challenging with multiple sources. The Super-UPS system is originally proposed for ac loads. It is also applicable for dc power distribution so that the PFC stage of the PSU can be removed.

D. Evaluation and Discussion

Nine data center power supply architectures from Figs. 2 to 8 are compared in Table I. The number of the conversion

stages is counted from the MV grid to the CPU. The ac and dc online UPS, ac PSU, and SST are considered to have two conversion stages. For efficiency comparison, we choose the overall efficiency from MV grid and the backup efficiency measured from the backup batteries. The efficiency calculation ends at the input of the POL converter. The input voltage of the PoL converter is assumed to be 48 V in each case for consistency. The losses from PDU and other mechanical parts are not included. The efficiency data of each conversion stage at 50% load are used considering the derating of power supplies. The overall efficiency is calculated by multiplying the efficiency of each stage.

Detailed converter specifications and efficiency calculation are given in the Appendix. The efficiency evaluation shows the power supply architectures with fewer conversion stages have higher efficiency in normal operation mode and backup mode. Traditional architectures with more conversion stages have relatively low efficiency. The performance evaluation does not cover the power density of all architectures due to the lack of dimension data. The volume estimation at 2.5-MW level in the Appendix shows that the Panama power supply offers about 30% space saving compared to the conventional ac UPS system with LFT. An SST prototype presented in [29] and [30] also demonstrates high-density power stage with 10-kV SiC device. Whereas, the overall power density of the current cascaded SST prototypes may be lower than the LFT solutions

due to the complex assembly structure, isolation distance for the semiconductor devices and circuit components, and extra protection circuitry for the cascaded modules. Further study is needed to increase the power density of SST.

Furthermore, the cable losses are also evaluated, which in fact, is important to the system efficiency and cost. Two scenarios are considered: power distribution at building level and power distribution among servers within a rack. In the first scenario, the ac and dc online UPS, Panama power supply, and SST are considered to be deployed close to the utility infrastructure. The corresponding distribution voltages of each architecture are listed in Table I. Using the same wiring configuration of [25] with Delta BL busbar (one conductor for each phase line and neutral line in ac distribution, two conductors for each rail in dc distribution), the cable loss ratio of the 220/380-Vac, 240-Vdc, 336-Vdc, and 400-Vdc distribution is 1:2.52:1.29:0.91. The boundary is 380 Vdc that the dc distribution is more efficient than the three-phase 220/380-Vac distribution. The cable loss at rack level with 240/336 and 400-Vdc is clearly lower than the single-phase 220-Vac distribution and the 48-Vdc distribution. Higher distribution voltage can significantly reduce the cable loss as the power of racks and servers increases. On the other hand, isolation and protection for high-voltage distribution require more attention and investment. In addition, SST can be deployed at each server room without the bulky MV to LV distribution transformer. Thereby the power is distributed at MV-level to each server room and the cable loss can be significantly reduced.

The reliability of the data center power supply system is highly related to the number of conversion stages, the location of backup battery, and the system-level redundancy. Similarly, fewer conversion stages in the power delivery path lead to higher reliability. The low voltage direct current (LVDC) system and the 48-V system have the distributed battery deployment because there are no centralized UPS or power supplies. Their reliability is considered to be better than other backup structure because the shelf-assembled batteries are physically closer to the load. On the other hand, the capacity and backup time of the LVDC system and the 48-V system are limited by the shelf space. Distribution of backup battery also requires more efforts on management and maintenance. The LVDC system also requires deeply customized PSU with build-in batteries.

The ac power distribution is mature and low cost. On the contrary, dc power distribution, especially at HVDC, has the issue of dc breaker, arc extinction, and protection. dc architectures are more suitable for well-designed and operated data centers. As the demand for decarbonization of Data centers increases, more renewable energy will be integrated into the power supply system. The concept of super-UPS will be gradually implemented in the data center.

III. AC–DC CONVERSION

The ac–dc conversions are needed in UPS, PSU, and SST. In most cases, ac–dc conversion is the first power conversion stage and delivers all power to servers. The efficiency and reliability of ac–dc conversion are critical to the entire data center

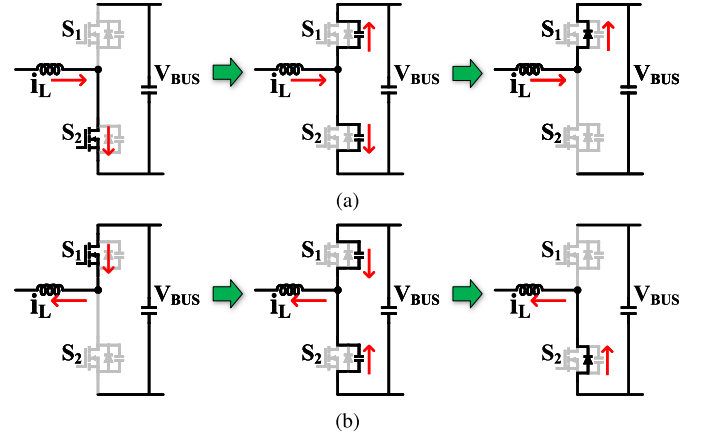


Fig. 9. Concept of TCM and CRM. (a) ZVS-ON of S_1 . (b) ZVS-ON of S_2 .

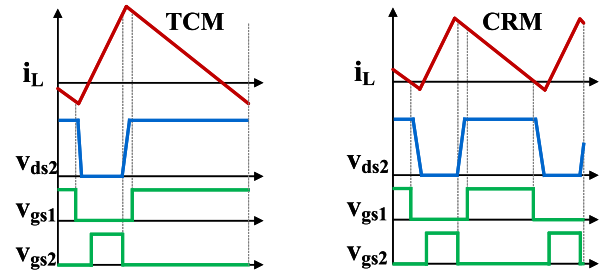


Fig. 10. Inductor current and MOSFET voltage under TCM and CRM operation.

power supply system. This section presents three updates to ac–dc converters in data centers.

A. TCM and CRM Converter

Triangle current mode (TCM) [36], [37], [38], [39] and critical conduction mode (CRM) [40], [41], [42], [43], [44] are two zero-voltage switching (ZVS) operation modes for ac–dc and dc–ac converters. One challenge to the conventional ac–dc converter is the switching loss especially from the reverse recovery of diodes. Wide bandgap (WBG) semiconductor devices like SiC and GaN device have excellent switching performance with almost zero reverse recovery current. Whereas, their turn-on loss is still much larger than the turn-off loss [45], [46], [47]. Besides, the fast-switching transition of WBG devices also causes EMI issues. Thereby, soft-switching, especially ZVS, is of great significance to WBG devices.

Figs. 9 and 10 show the concept of TCM and CRM operation in one phase leg and the related inductor current and metal oxide silicon field effect transistor (MOSFET) voltage. The key idea is to avoid the diode freewheeling by changing the inductor current direction when the switch is turned off. With TCM operation, S_1 in Fig. 9(b) is turned off when the inductor current decreases to a negative boundary. The current direction of S_1 is from drain to source so that there is no diode freewheeling. The output capacitance of S_1 and S_2 is charged and discharged by i_L . Then the body-diode of S_2 conducts after its output capacitance is discharged to zero. The negative current of inductor is necessary for ZVS even the grid instant

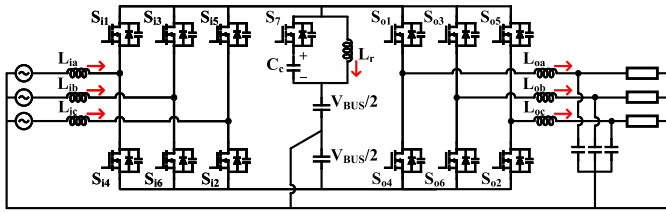


Fig. 11. Topology of three-phase four-wire ZVS BTB converter [56].

current is zero. With CRM operation, S_1 is turned off when the inductor current decreases to zero. The output capacitance of switch starts to resonate with the filter inductor. The ZVS condition of CRM is that the dc bus voltage is higher than twice of the instantaneous grid voltage.

The TCM and CRM both introduce high current ripple to semiconductor devices and filter inductors, leading to a 15% increment on root-mean-square (RMS) current compared to the continuous conduction mode (CCM) converter. Another challenge for TCM and CRM operation is the variation of switching frequency, which may be complicated for control and filter design. Modifications of TCM and CRM have been proposed to narrow the switching frequency range including adjusting the current boundary [37] and combining CRM, TCM with discontinuous conduction mode (DCM) [44]. Despite the challenges, TCM and CRM converters are still attracting research interest for their effectiveness on ZVS and excellent efficiency and power density. TCM and CRM technology for the single-phase PFC rectifier in server PSU have been investigated in [36], [38], [39], [40], [41], and [42]. With ZVS and WBG device, the switching frequency of the TCM and CRM PFC converters is pushed to MHz range with a peak efficiency around 99%. A single-stage TCM PFC converter with 10-kV SiC device is proposed for the MV grid interface [30]. The switching frequency is between 35 and 75 kHz with direct conversion from 3.8 kVac to 7 kVdc and the full-load efficiency reaches 99.1% at 25 kW, proving the effectiveness of the TCM technology. The TCM and CRM operation with nonlinear load, which is typical in UPS applications, still requires investigation.

B. ZVS BTB Converter

Another soft-switching approach is to use the auxiliary circuit to resonate the entire dc link voltage to zero before the switching commutation. This soft-switching concept is used in resonant dc-link (RDCL) inverter [48], [49], which is the pioneer and has a significant impact on this area. It originally uses discrete pulse modulation (DPM) with variable switching frequency, resulting in sub-harmonics in the grid-side current. pulsewidth modulation (PWM) controls for the resonant dc link concept also have been investigated and applied to single/three-phase ac-dc/dc-ac converters [50], [51], [52], [53], [54], [55]. In these topologies, one auxiliary circuit covers the ZVS for one converter. A major feature of this ZVS technology is that the filter inductor is in continuous conduction mode and the conduction loss of device is not increased.

Given the fact that the ac-dc stage and the dc-ac stage of the ac online UPS share a common dc bus, the resonance between the auxiliary circuit and the dc link potentially can create ZVS for both stages. A three-phase four-wire ZVS BTB converter is proposed in [56] and the topology is shown in Fig. 11. The auxiliary circuit is inserted between the bus capacitors and the dc link of two three-phase converters. During most of the switching cycle, the auxiliary switch S_7 is conducting and the dc link voltage V_{link} is the summation of the bus voltage V_{BUS} and the clamping voltage V_{Cc} . Prior to any high loss commutation (from the diode or body-diode of a switch to its complement switch in the same phase leg) of three-phase converters, S_7 is turned off to initiate the resonance between the resonant inductor L_r and the equivalent capacitance (the output capacitance of the semiconductor power device and optional external capacitors) of the dc link. The targeting switch can achieve ZVS-ON after the entire dc link voltage decreases to zero.

The switching action of S_7 must be timed at every commutation with higher switching loss (Type-2 commutation [55], [57]) in two three-phase converters. As shown in Fig. 12(a), S_7 needs to switch six times in each switching cycle with conventional sine pulse width modulation (SPWM), resulting in complicated control and extra circulation loss in the circuit. An edge-aligned PWM (EA-PWM) [57] has been proposed to synchronize all commutations with higher switching loss and reduce the switching frequency of S_7 . As shown in Fig. 12(b), the switching frequency of S_7 is the same as the switching frequency of all other six switches. The ZVS topology and EA-PWM are further extended to more complex system with both ac and dc conversions [58]. The fundamental and detailed implementation of the ZVS technology is comprehensively introduced in [59]. It is potentially applicable for the renewable integration of data center power supply systems.

C. Three-Level BTB Converter With Hybrid Module

Three-level topologies have been widely used in PV inverters, high-power UPS, motors drives, and so on. Compared to two-level topologies, three-level topologies have lower voltage stress on semiconductor devices, smaller dv/dt and lower switching harmonics [60], [61]. Fig. 13 shows the topology of a T-type three-level BTB converter. All the vertical switches need to block the entire dc bus voltage V_{bus} , while the voltage stress of horizontal switch is only $V_{bus}/2$. Compared to the neutral point clamped (NPC) three-level converter, the T-type converter has lower conduction loss and higher switching loss. Thereby it is suitable for low-voltage high-power application such as UPS.

The reverse recovery loss of the anti-parallel Silicon diode of insulated gate bipolar transistor (IGBT) has significant impact on the system efficiency. Replacing the anti-parallel Silicon diode by SiC Schottky barrier diode (SBD) becomes attractive because: 1) SiC diodes have almost zero reverse recovery loss; 2) the main circuit and gate drivers do not need to be changed; and 3) the cost is lower than all SiC solution. Device manufacturers have released hybrid half-bridge module [62] and hybrid three-level NPC module [63].

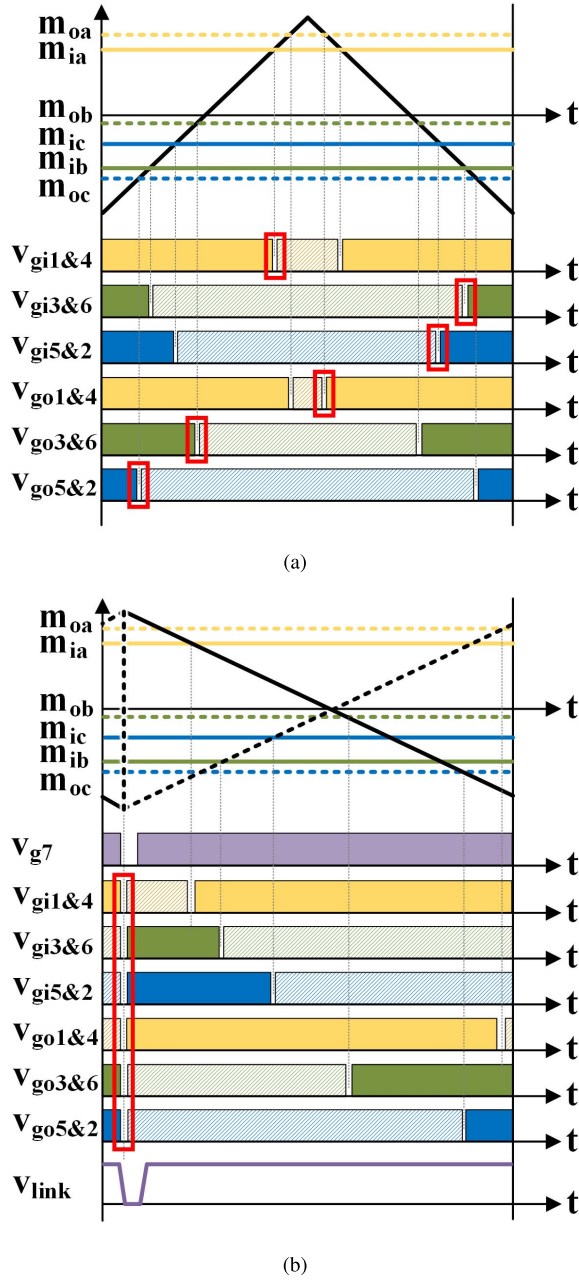


Fig. 12. Modulation waves, carrier waves, and gate signals of the conventional SPWM and the EA-PWM. The commutations with diode reverse recovery are marked by red rectangles [58]. (a) Conventional SPWM. (b) EA-PWM.

In the T-type three-level BTB converter, reverse recovery of the anti-parallel diode exists in the vertical switches of the rectifier and in the horizontal switches of the inverter with unity power factor (PF) operation. Fuji Electric released a 225–333-kVA T-type three-level UPS with hybrid SiC half-bridge module as the vertical switches in the rectifier [64]. Due to the unavailability of the hybrid SiC module with bidirectional configuration, the inverter still uses all Silicon devices. The efficiency is 96% at 25% load and 97.3% at full load.

A T-type three-level BTB converter with hybrid modules in both the rectifier and the inverter is researched in [65]. The

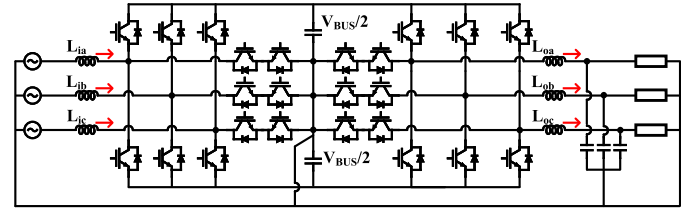


Fig. 13. T-type three-level BTB converter with IGBT.

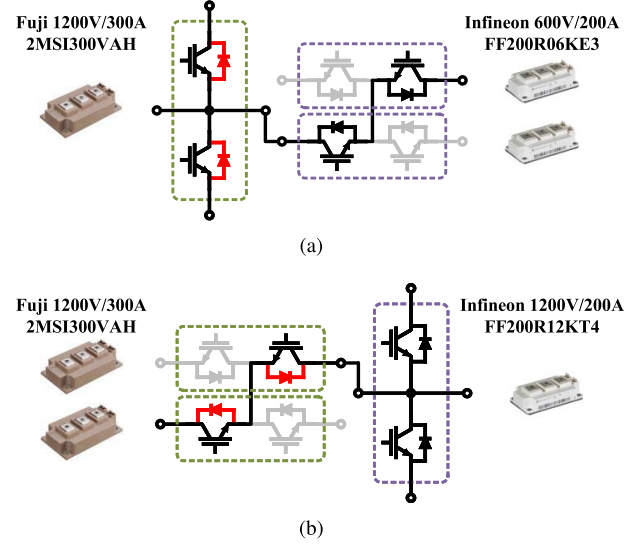


Fig. 14. Hybrid module configuration of a T-type three-level BTB converter. SiC SBDs are marked in red [65]. (a) One rectifier phase. (b) One inverter phase.

module configuration is shown in Fig. 14. A hybrid half-bridge module with Silicon IGBT and SiC SBD is used as the vertical switches in the rectifier phase. The bidirectional horizontal switch is implemented by two half-bridge IGBT modules. Two hybrid half-bridge modules form the bidirectional horizontal switch of the inverter phase and a 1200 V half-bridge module is used as the vertical switches. This configuration is suitable for unity PF operation. The full load efficiency at 45 kW is 97.0% at 7.2 kHz, and 96.6% at 15 kHz, which is 0.3% and 0.6% higher than the converter with all Silicon devices. The efficiency improvement also shows the effectiveness of the hybrid module solution for higher switching frequency. Furthermore, the efficiency will be higher if the vertical modules of the inverter are 600-V devices.

D. Comparison of BTB Converters

Table II lists the key specifications of the ZVS BTB converter prototypes and the three-level BTB converter with hybrid module that all developed by the authors of this article. The detailed design and test results are presented in [65] and [56]. The ZVS two-level BTB prototype and the hard-switching two-level BTB prototype have the same operation condition, same SiC MOSFET, and the same circuit layout except the auxiliary circuit. The efficiency of the ZVS BTB converter is 3% higher at both half-load and full-load. The efficiency above 97% at 150 kHz also indicates the high power

TABLE II
COMPARISON OF BTB CONVERTERS IN AC ONLINE UPS

Topology (Product)	Wiring	Power	V_{ac}	V_{bus}	Rectifier Device	Inverter Device	Switch. Freq.	50% Load Eff.	100% Load Eff.
ZVS 2-Level [56]	3P4W	10 kVA	220 V	700 V	SiC-MOS	SiC-MOS	150 kHz	97.5%	97.1%
Hard-Switching 2-Level [56]	3P4W	10 kVA	220 V	700 V	SiC-MOS	SiC-MOS	150 kHz	94.3%	94%
ZVS 2-Level [65]	3P4W	50 kVA	220 V	700 V	Si-IGBT	Si-IGBT	15 kHz	96.46%	96.18%
Hard-Switching 2-Level [65]	3P4W	50 kVA	220 V	700 V	Si-IGBT	Si-IGBT	15 kHz	94.3%	93.95%
Hybrid T-Type [65]	3P4W	50 kVA	220 V	700 V	Si-IGBT + SiC-SBD	Si-IGBT + SiC-SBD	7.2 kHz	97.2%	97%
							15 kHz	96.8%	96.6%
ABB PowerScale [66]	3P4W	10 kVA	230 V	NA	NA	NA	NA	Peak eff. = 95.5%	
Delta HPH-40K [67]	3P4W	40 kVA	230 V	NA	NA	NA	NA	Peak eff. = 96%	

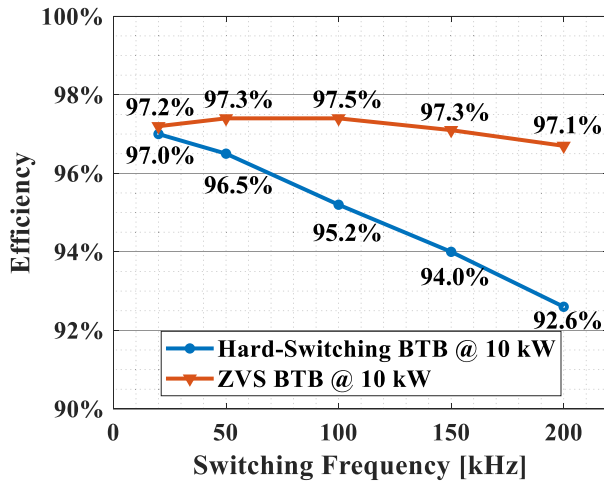


Fig. 15. Efficiency comparison between the 10-kW ZVS 2-level SiC BTB converter and the hard-switching 2-level SiC BTB converter at different switching frequencies [65].

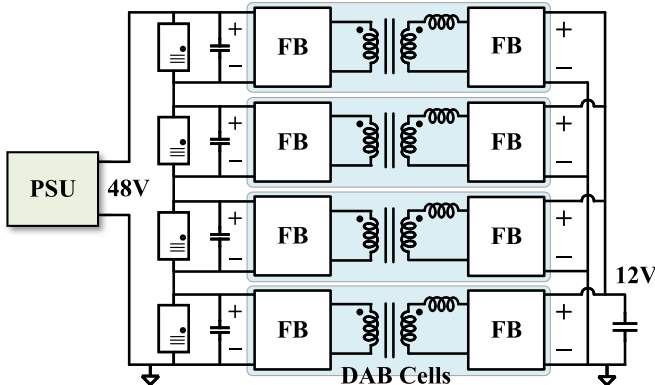


Fig. 16. DPP power supply for four 12-V stacked servers [83].

density of the ZVS BTB converter. Similar results are found in a high-power ZVS BTB prototype with IGBT module. Both the ZVS two-level BTB converter and the T-type three-level BTB converter with hybrid SiC + Si device [65] have higher

efficiency than the commercial products with similar power level [66], [67].

Fig. 15 compares the efficiency of the two-level SiC BTB converter with ZVS and hard-switching. The efficiency with ZVS drops slightly as the switching frequency increases from 20 to 200 kHz. Thereby the ZVS BTB topology is suitable for high-frequency operation up to hundreds of kHz with WBG devices. The hybrid SiC + Si solution is more cost effective than the full-SiC solution. Its performance can be further improved if SiC and Si devices are packed in the same module according to the circuit configuration.

IV. DC-DC CONVERSION

The dc-dc conversions in data centers include the isolated second stage of dc online UPS and PSU, the bidirectional converters for backup batteries and the PoL converters. LLC Resonant topologies with inherent soft-switching have been widely applied to the isolated dc-dc conversion from the HVDC bus to the low-voltage output at 12 V or 48 V [68], [69], [70], [71], [72], [73]. WBG devices, soft-switching, and novel transformer design [70], [71], [72], [73], [74], [75] enable high efficiency and power density with the switching frequency up to MHz range. Recent publications also show the effectiveness of modular topologies with series-input parallel-output structure for the HVDC to 12 V or 48 V conversion [76], [77], in which the high-voltage stress can be shared by modular cells with low-voltage device and the voltage second of transformer is also reduced. The reported peak efficiency is 98.6%–99% with a 48-V output [70], [71], [72] and this number has reached 98.4% with a 12-V output [76].

This section will introduce an efficient power conversion concept called the differential power processing (DPP), and explore its potential applications in data centers. In addition, recent developments on the high conversion ratio PoL converters are also included.

A. Differential Power Processing in Data Centers

The concept of differential power processing was initially applied to battery systems [78], [79], in which the power

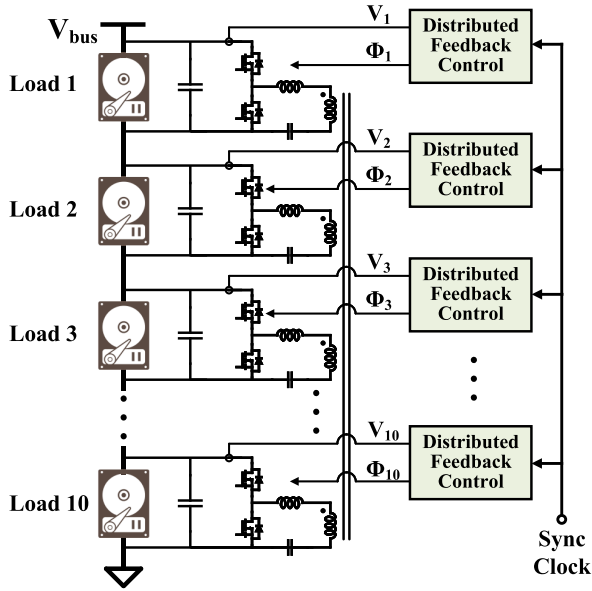


Fig. 17. MAB DPP power supply for storage server [84].

converters only process the charge difference among series battery cells for equalization. Similar DPP idea is later used to deal with the power mismatch from partial shading of PV strings [80], [81], [82]. Recently, the DPP architecture is also extended to data center applications to power the stacked servers [83], stacked hard disk drives (HDD) [84], and stacked processors [85]. The nature of DPP enables high efficiency and high power density in these particular systems. Detailed review and classification of DPP architectures can be reached in [86] and [87].

Fig. 16 illustrates the DPP power supply for stacked servers proposed in [83]. Four 12-V servers are connected in series and powered by the 48-V bus from the PSU. For the stacked servers, unequal power consumption leads to unbalanced input voltage of each server. Then the DPP converter is needed to draw energy from the port with less server power consumption and compensate energy to the port with higher power consumption. The DPP converter is made up by four dual-active-bridge (DAB) cells and a buffer capacitor, in which the differential power is exchanged through two DAB cells. It can achieve a peak efficiency of 98.7% owing to the small portion of differential power. Furthermore, the DPP converter does not need to be designed for the total power of four servers, so that the cost and size of DPP converter are also reduced.

A ten-port multi-active-bridge (MAB) converter is designed as the DPP converter for storage servers [84]. The system schematic is shown in Fig. 17. The 5-V HDDs in the storage server are grouped in series and powered by a 50-V bus. The MAB converter with ten half-bridge cells and one multiwinding single-core transformer interconnects all the ten groups. The operation concept is similar to the DPP system in Fig. 16. The differential power between two arbitrary ports only needs to be processed by two half-bridge cells with less power loss. The single core design further improves the power density. The reported system efficiency is 99.77% with a power density of 700 W/in³.

One limitation of the above DPP systems is that the bus voltage is only determined by the PSU. That means any voltage fluctuation on the bus will directly change the voltage of each load. To address this issue, a buck-type series voltage compensator [88] is employed between the ten-port MAB DPP converter and the 50-V bus to absorb the bus voltage variation. Whereas the overall efficiency drops by 1%–3% with a 5-V difference. The floating ground in the DPP system also requires differential signal transmission or even optical communication. Another concern is that the output of DPP system is not galvanically isolated from its input, which can cause potential common-mode and safety issues especially at high-voltage application such as 400–48-V conversion. In the authors' opinion, the applicable applications of the DPP architecture are battery equalizer of BBU and the interface between the new 48-V system and the legacy 12-V servers.

B. 48-V Point-of-Load Converters

As the last stage to power the CPU, PoL converters face the challenges of high conversion ratio, high output current, and high load transient. In the 12-V systems, typical PoL solutions utilize the single-stage multiphase buck topology as the voltage regulator module (VRM) [89]. Due to the high output current, extreme duty ratio at 1 V output and the hard-switching of buck converter, the peak efficiency of 12–1 V conversion with DrMOS just reaches 90%, making it the bottleneck of the overall efficiency of the data center power supply system. Now the data center industry is undergoing the transition to the 48-V system for lower power distribution loss. 48-V PoL solutions usually cascade two conversion stages with an intermediate bus to extend the conversion ratio [90]. The intermediate bus architectures (IBAs) with regulated bus and unregulated bus are systematically reviewed and compared in [91]. Besides, two stages can be stacked as ISOP. Based on how two stages are connected, we classify them into four architectures shown in Fig. 18.

Fig. 18(a) shows the two-stage IBA that adds an intermediate bus converter in front of the VRM. dc transformer (DCX) topologies [92], [93], [94], [95] and switched-capacitor (SC)-based topologies [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106] are used for the bus converter because voltage regulation is not required in this stage. The mature control of VRMs can be directly applied, enabling the fast migration to the 48-V system. Since both DCX and SC topologies can achieve high efficiency up to 99%, the efficiency of the VRM is still the bottleneck of the overall efficiency of two-stage IBA. Reducing the IB voltage becomes attractive because it can reduce the VRM's switching loss and improve the dynamic performance at higher switching frequency [104], [105], [106]. Intel has demonstrated a 5-V VRM with low-voltage GaN nMOS transistors [107]. The reported full load efficiency is 88.8% at 1 V/32 A and 3.1 MHz.

The two-stage IBA with preregulation shown in Fig. 18(b) is patented by Vicor [108], [109], in which the first stage is a buck-boost converter and the second stage is a LLC-DCX. The intermediate bus voltage is 48 V or 32 V with the 48-V input.

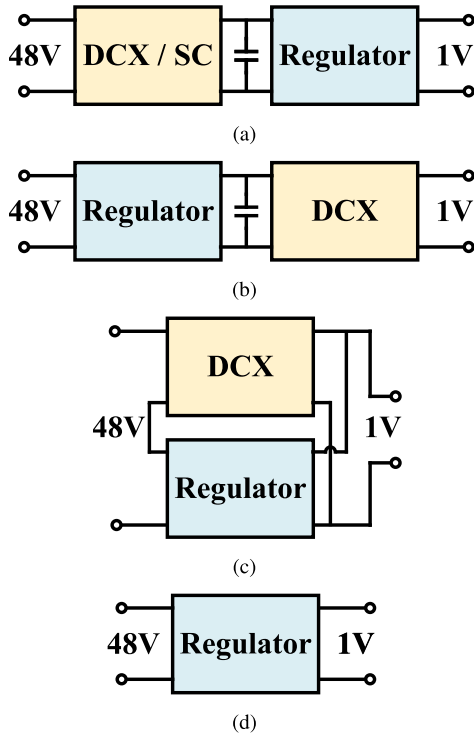


Fig. 18. Four 48-V PoL architectures. (a) Two-stage IBA. (b) Two-stage IBA with Preregulation. (c) ISOP. (d) Single-stage.

Therefore, the inductor of the buck-boost stage has both low dc bias and ripple current, and significantly reduced size. Vicor's 48-V PoL solution has the industry leading power density. Its dynamic response may be slow due to long power path and the inherent nonlinear resonant behavior. Fig. 18(c) shows the ISOP architecture [110], in which the input terminals of the DCX and regulator are connected in series and their outputs are connected parallel. The power rating of the regulator is only part of total power and the conversion loss is reduced. On the other hand, its regulation ability may be limited by the power rating of the regulator.

In fact, single-stage isolated topologies with current doubler are the earliest work on 48-V PoL applications [111], [112], [113], [114]. Recently they have raised research interest again because of the simple circuit topology [115], [116], [117], [118], [119]. Another single-stage approach combines the SC concept with inductor to provide both output regulation and high step-down conversion [120], [121], [122], [123], [124]. The dynamic performance of the single-stage isolated/SC solutions requires further investigation because their maximum duty ratio is limited below 50%. Similar SC-based regulation topologies are also applied to two-stage architectures. Fig. 19 shows a four-phase buck converter and a four-phase series-capacitor buck (SCB) converter [125], [126]. Four high-side switches of the series-capacitor buck converter are connected in series and the intermediate bus voltage V_{IB} is equally shared by four switching cells. Thereby, the effective input voltage of each phase is only $V_{IB}/4$, which significantly reduces the voltage stress and switching loss. The series-capacitor buck topology is also applied to the two-stage solutions as

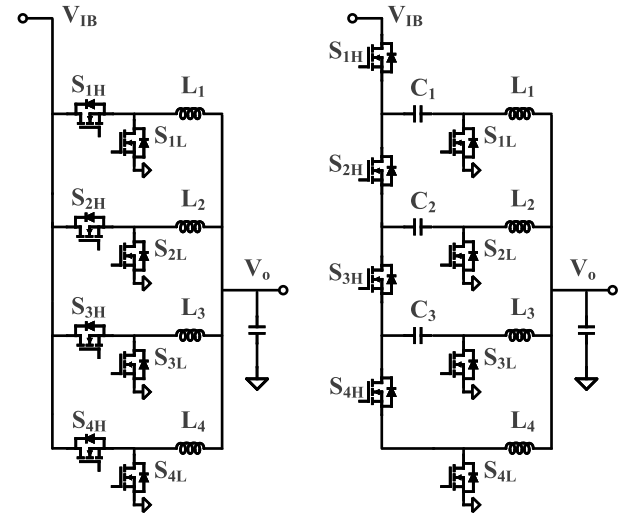


Fig. 19. Four-phase buck converter and the four-phase series-capacitor buck converter.

the regulation stage with high efficiency and power density [127], [128].

C. Comparison of 48-V PoL Converters

Table III compares a few key metrics of the recent 48-V to point-of-load voltage regulator designs. The 48-V PoL topologies are categorized as four groups: two-stage IBA, two-stage preregulation, ISOP, and single-stage. The comparison shows the single-stage solutions including the transformer-based topologies and SC-based topologies have relatively high efficiency. In the authors' opinion, two-stage and single-stage solutions are both needed in the future. Two-stage IBA is more suitable if multiple low-voltage rails (e.g., 1, 1.8, 3.3 V) are needed. The intermediate bus voltage design to balance the overall efficiency, power density, and the dynamic performance will be one focus. The single-stage solutions can be applied to system with less complexity but requires high current. Meanwhile the dynamic performance and the transient control of the single-stage solutions require more investigation.

V. SUMMARY

This article presents an architecture-level review of data center power supply systems first, including the ac architecture, dc architectures, and hybrid architectures. There is a trend to reduce the power conversion stages to improve the overall efficiency and reliability. Therefore, dc architectures are promising because the dc-ac stage in UPS and the ac-dc stage in PSU can be eliminated. Recent developments on SST also enable the direct interface to the MV grid without the bulky LFT. Renewable integration is a new demand, that requires the innovations on circuit topology and control.

The recent developments on the converter-level in data center power supply systems are also introduced in this article, including the ZVS technologies with WBG devices and a low-cost alternative with hybrid SiC SBD and IGBT module for ac-dc/dc-ac conversions in data centers, the server dc

TABLE III
PERFORMANCE COMPARISON OF 48-V POINT-OF-LOAD VOLTAGE REGULATOR DESIGNS

Archit.	Topology	V_{in} to V_o	Max. I_o	50% Load Eff.	100% Load Eff.	Height	Current Density	Switch. Freq. [†]	Including Gate Loss & Size
Two-Stage IBA	LLC+Buck [93]	48-6-1.8 V	530 A	93.5% [‡]	89% [‡]	NA	NA	600 kHz	NA
	LLC+Buck (MPS) [95]	48-4.8-0.8 V	333 A	NA	86.4%	NA	0.17 A/mm ²	NA	Yes
	SC+Buck (ADI) [103]	48-12-1 V	50 A	90.7%	88.1%	7.72 mm	0.064 A/mm ²	350 kHz	Yes
	SC+SCB [127]	48-24-1 V	150 A	90.5%	86.2%	20 mm	0.346 A/mm ²	417 kHz	No
	SC+Buck [105]*	48-8-1 V	780 A	86.5%	79.2%	16.65 mm	1 A/mm ²	1 MHz	No
	SC+SCB [128]*	48-24-1 V	640 A	91.8%	84.4%	11 mm	0.311 A/mm ²	417 kHz	No
Two-Stage Pre.	Buck-Boost + LLC (Vicor) [109]	48-48-1 V	120 A	Est. eff. = 90.2% [‡]		6.73 mm	0.202 A/mm ²	1.025 MHz	Yes
ISOP	LLC+Buck (Sigma) [110]	48-1 V	80 A	94%	92.5%	4 mm	0.127 A/mm ²	600 kHz	Eff.: No Density: Yes
Single-Stage	Current Doubler (Bel Power) [116]	48-1 V	70 A	92%	90.5%	12.7 mm	0.184 A/mm ²	242 kHz	Yes
	Current Doubler [119]	48-1.8 V	160 A	92.7%	90%	4.45 mm	0.152 A/mm ²	600 kHz	Yes
	12-Level SCB On-Chip [122]	48-1 V	8 A	86%	76%	2.6 mm	0.031A/mm ²	2.5 MHz	Yes
	Hybrid-SC [123]	48-1 V	270 A	91.5%	87.7%	6.45 mm	0.142A/mm ²	280 kHz	Yes

[†] The switching frequency of the voltage regulation stage.

[‡] Calculated by multiplying the efficiencies of two stages.

* These solutions are also called the merged-two-stage architecture with small decoupling capacitor.

power supply with DPP concept and 48-V PoL converters. These emerging technologies all target lower device stress and faster operating frequency to improve the efficiency, power density, and dynamic performance of converters.

Given the fact that the power consumption of data centers is huge and continues to grow. The importance of the efficiency of data center power supply system is obvious. Energy saving and decarbonization in data centers requires efforts on both system architecture and every power conversion stage from the grid edge to onboard processors.

APPENDIX

SPECIFICATIONS AND PERFORMANCE EVALUATION OF DATA CENTER POWER ARCHITECTURE

One fact is that the power conversion in the data center is more centralized at the grid edge (e.g., high-power LFT and UPS) and more distributed at the load side. At each conversion stage, we consider there are multiple modules in parallel. e.g., one LFT, several UPS, hundreds, or thousands of PSU. Thereby, the overall efficiency of the data center power supply system can be evaluated by multiplying the efficiency of each converter and the difference of the converter's power rating does not affect the calculation. The efficiency data with 50% load is selected considering the load utilization and design overhead.

A 99% efficiency is reasonable for the MW-level LFT. Table IV lists the specifications of four high-power ac online

UPS with leading performance. We choose the number of 97% for the general case with Silicon device. The efficiencies of the three-phase rectifier and three-phase inverter in the ac online UPS are both set to 98.5% with the symmetrical structure. The efficiency of the two-stage 220-Vac to 48-Vdc PSU is set to be 97% at the power level of 3 kW [129], [130]. The normal operation efficiency and the backup efficiency of the power architecture with ac online UPS are

$$\eta_{ac_normal} = 99\% \times 97\% \times 97\% = 93.15\%$$

$$\eta_{ac_back} = 98.5\% \times 97\% = 95.55\%. \quad (1)$$

We assume the dc online UPS has the same efficiency as the ac online UPS. The single-stage isolated dc PSU faces the input voltages of 240, 336, and 400 Vdc. Considering the distribution of switching loss and conduction loss with different voltages, we assume a 98% efficiency for all three cases. The efficiency data of first two HVDC architectures in Fig. 3 can be obtained.

The conventional HVDC architecture

$$\eta_{HVDC1_normal} = 99\% \times 97\% \times 97\% = 93.15\%$$

$$\eta_{HVDC1_back} = 97\%. \quad (2)$$

The HVDC architecture with dc PSU

$$\eta_{HVDC2_normal} = 99\% \times 97\% \times 98\% = 94.11\%$$

$$\eta_{HVDC2_back} = 98\%. \quad (3)$$

TABLE IV
SPECIFICATIONS OF AC ONLINE UPS

Product	Wiring	Power	Voltage	50% Load Eff.	100% Load Eff.	Volume	Weight	Remarks
DPA 500 [13]	3P4W & 3P3W	300 kVA	230 (400) V	NA	>96%	3.21 m ³	1225 kg	ABB
MegaFlex [13]		1.25 MVA		97.4%	>97%	6.09 m ³	2595 kg	
G9000 [14]	3P3W	500 kVA	480 V	97.1%	97.0%	2.99 m ³	1524 kg	Toshiba
G2020 [15]	3P3W	500 kVA	480 V	98.2%	97.6%	2.61 m ³	1250 kg	Toshiba SiC device
	3P3W	750 kVA	480 V	98.0%	97.5%	3.61 m ³	1810 kg	

TABLE V
SPECIFICATIONS OF PANAMA POWER SUPPLY AND TWO SST PROTOTYPES

Topology (Product)	Power	V_{high}	V_{low}	50% Load Eff.	100% Load Eff.	Volume	Weight	Remarks
Panama [23]	2.5 MW	10 kVac	240/336 V	Peak eff. = 97.5%		15.36 m ³	NA	Transformer: 4 racks Rectifier & PDU: 8 racks
SST [29] + [30]	25 kW	3.8 kVac	400 V	97.8%	98.1%	0.0116 m ³	NA	Single-phase AC input Volume of power stage
SST [32]	400 kW	4.8/13.2 kVac	1050 V	98.5%	98%	8.463 m ³	3000 kg	Designed for EV charger
SST [33]	1 MW	10 kVac	400-800 V	97.9%	97.5%	NA	NA	Designed for PV Eff. measured at 800-Vdc

The efficiency calculation of the simplified LVDC architecture can use the data of ac PSU and dc PSU to 400-Vdc SST

$$\begin{aligned}\eta_{LVDC_normal} &= 99\% \times 97\% = 96.03\% \\ \eta_{LVDC_back} &= 98\%.\end{aligned}\quad (4)$$

The efficiency calculation of the 48-V rack system is similar. The 48-V BBU efficiency is set to 98% [22]

$$\begin{aligned}\eta_{48\text{ V_normal}} &= 99\% \times 97\% = 96.03\% \\ \eta_{48\text{ V_back}} &= 98\%.\end{aligned}\quad (5)$$

The specifications of a 2.5-MW Panama power supply are listed in Table V. The data source of the Panama power supply [23] does not give the exact dimension but mentions that a standard rectifier rack and a PDU rack can support the power up to 720 kW, the phase shift transformer together with the input power distribution circuit occupy the space of 4 standard racks. The overall volume of the 2.5-MW system is estimated as 15.36 m³ (12 racks). If we use the same transformer space with two 1.25-MW MegaFlex and four more additional PDU racks, the overall volume of this 2.5-MW ac system is 22.42 m³. This comparison shows the high power density of the Panama power supply. The efficiency of the Panama power supply is considered to be 97.5%, and the overall system efficiency and backup efficiency are

$$\begin{aligned}\eta_{Panama_normal} &= 97.5\% \times 98\% = 95.55\% \\ \eta_{Panama_back} &= 98\%.\end{aligned}\quad (6)$$

Table V also shows the specifications of three SST prototypes. Considering the different input-output voltages, a 97.5% efficiency is used for the calculation with for a 10-kVac

$$\begin{aligned}\eta_{SST_normal} &= 97.5\% \times 98\% = 95.55\% \\ \eta_{SST_back} &= 98\%.\end{aligned}\quad (7)$$

The power density of SST is not compared due to the lack of fully commercialized products. [33] shows the 1-MW SST prototype is assembled in a standard container. The volume of the container is typically around 25 m³ and larger than the Panama power supply and the mature online UPS system. Further study is needed to improve the power density of SST.

In the ac + dc hybrid system we assume the input power is equally shared and the overall efficiency is the weighted average efficiency of two paths

$$\begin{aligned}\eta_{Hybrid_normal} &= 99\% \times (0.5 \times 97\% + 0.5 \times 97\% \times 98\%) \\ &= 95.07\% \\ \eta_{Hybrid_back} &= 98\%.\end{aligned}\quad (8)$$

The super-UPS system is implemented by 100-kW bi-directional dc-ac modules and 30-kW bi-directional dc-dc modules, both are non-isolated [35]. The dc bus voltage is ± 375 V and the input range of the dc-dc module is 200–400 V. We also assume the dc-ac modules and dc-dc modules in the super-UPS system all have a 98.5% efficiency so that power distribution in the backup mode does not affect the efficiency. The efficiency of the super-UPS system can be calculated by

$$\begin{aligned}\eta_{Super_normal} &= 99\% \times 97\% \times 97\% = 93.15\% \\ \eta_{Super_back} &= 98.5\% \times 97\% = 95.55\%.\end{aligned}\quad (9)$$

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